

Machine Learning Operations (MLOps): Overview, Definition, and Architecture

Dominik Kreuzberger
KIT

Germany
dominik.kreuzberger@alumni.kit.edu

Niklas Kühl
KIT

Germany
kuehl@kit.edu

Sebastian Hirschl
IBM[†]

Germany
sebastian.hirschl@de.ibm.com

ABSTRACT

The final goal of all industrial machine learning (ML) projects is to develop ML products and rapidly bring them into production. However, it is highly challenging to automate and operationalize ML products and thus many ML endeavors fail to deliver on their expectations. The paradigm of Machine Learning Operations (MLOps) addresses this issue. MLOps includes several aspects, such as best practices, sets of concepts, and development culture. However, MLOps is still a vague term and its consequences for researchers and professionals are ambiguous. To address this gap, we conduct mixed-method research, including a literature review, a tool review, and expert interviews. As a result of these investigations, we provide an aggregated overview of the necessary principles, components, and roles, as well as the associated architecture and workflows. Furthermore, we furnish a definition of MLOps and highlight open challenges in the field. Finally, this work provides guidance for ML researchers and practitioners who want to automate and operate their ML products with a designated set of technologies.

KEYWORDS

CI/CD, DevOps, Machine Learning, MLOps, Operations, Workflow Orchestration

1 Introduction

Machine Learning (ML) has become an important technique to leverage the potential of data and allows businesses to be more innovative [1], efficient [13], and sustainable [22]. However, the success of many productive ML applications in real-world settings falls short of expectations [21]. A large number of ML projects fail—with many ML proofs of concept never progressing as far as production [30]. From a research perspective, this does not come as a surprise as the ML community has focused extensively on the building of ML models, but not on (a) building production-ready ML products and (b) providing the necessary coordination of the resulting, often complex ML system components and infrastructure, including the roles required to automate and operate an ML system in a real-world setting [35]. For instance, in many industrial applications, data scientists still manage ML workflows manually

to a great extent, resulting in many issues during the operations of the respective ML solution [26].

To address these issues, the goal of this work is to examine how manual ML processes can be automated and operationalized so that more ML proofs of concept can be brought into production. In this work, we explore the emerging ML engineering practice “Machine Learning Operations”—MLOps for short—precisely addressing the issue of designing and maintaining productive ML. We take a holistic perspective to gain a common understanding of the involved components, principles, roles, and architectures. While existing research sheds some light on various specific aspects of MLOps, a holistic conceptualization, generalization, and clarification of ML systems design are still missing. Different perspectives and conceptions of the term “MLOps” might lead to misunderstandings and miscommunication, which, in turn, can lead to errors in the overall setup of the entire ML system. Thus, we ask the research question:

RQ: What is MLOps?

To answer that question, we conduct a mixed-method research endeavor to (a) identify important principles of MLOps, (b) carve out functional core components, (c) highlight the roles necessary to successfully implement MLOps, and (d) derive a general architecture for ML systems design. In combination, these insights result in a definition of MLOps, which contributes to a common understanding of the term and related concepts.

In so doing, we hope to positively impact academic and practical discussions by providing clear guidelines for professionals and researchers alike with precise responsibilities. These insights can assist in allowing more proofs of concept to make it into production by having fewer errors in the system’s design and, finally, enabling more robust predictions in real-world environments.

The remainder of this work is structured as follows. We will first elaborate on the necessary foundations and related work in the field. Next, we will give an overview of the utilized methodology, consisting of a literature review, a tool review, and an interview study. We then present the insights derived from the application of the methodology and conceptualize these by providing a unifying definition. We conclude the paper with a short summary, limitations, and outlook.

[†] This paper does not represent an official IBM statement